# § 4-3 prime numbers

$b \mid a$

(a) $1 \mid a$ , $a \mid 0$

(b) $a \mid b \land b \mid c \Rightarrow a \mid c$

(c) $a \mid b \land b \mid a \Rightarrow a = \pm b$

$a \mid b$

⋮

(f) $(a \mid b) \land (a \mid c) \Rightarrow a \mid (bx + cy)$

Example:

$a . b \in \mathbb{Z}$ such that $(17 \mid 2a+3b)$

to prove $17 \mid (9a+5b)$

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<pf>

$17 \mid 2a+3b$

$17 \mid (2a+3b)(-4)$

$17 \mid 17$

$17 \mid 17(a+b)$

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$\therefore 17 \mid (\quad) \times 1 + (\quad) \times 1$

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integer $k$ exactly two disvor 質數

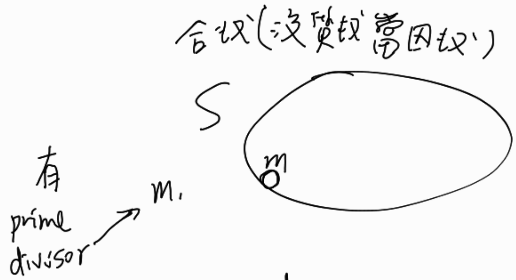
composite number 合數

有 ? 數

2 . 3 . 5 . 7 . 11 . 13 . 17 . 19 . 23 . 29 . 31

<pf> Lemma If $n \in \mathbb{Z}^+$, $n$ is composite then there exist a prime

s.t. $p \mid n$ 2 3 5 7




方法

$\boxed{\neg P \to F \quad 假設p無質因數}$

$S$ be the set of all composite integers have no prime divisor.

沒有人沒有質因數 應 $S = \phi$

If $S \neq \phi$ $\left\{ \begin{array}{l} S \subseteq \mathbb{Z}^+ \\ S \neq \phi \end{array} \right\}$ By the well-ordering property

沒有沒有人沒有質因數 應 $S = \phi$

a least element $m$

m:合
有質因數

2

$\because$ 合數 $m = m_1 \cdot m_2$

$\because$ 不是質數 $m_1 \neq 1$ $\therefore m_1 < m$
$m_2 \neq 1$ $m_2 < m$
合

3

$m_1$, $m_2$: 合數有質數當因數

$m \in S$ $m_1$: 不是質數
$m_2$

(原本設m是沒有 prime divisor)

$p \mid m_1$ $m_1 \mid m$ $\therefore p \mid m$ 結果應該要是可整除

✗

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Thm infinite prime

<pf> If finite $p_1$ $p_2$ $p_3$ $\cdots$ $p_k$ finite list

證明無限多個質數
設有限 最後要 ✗

Let $B = p_1 \times p_2 \times p_3 \times \cdots \times p_k + 1$ 設有限

$B > p_i$

$\therefore B$ $\therefore$ composite
$p_i \mid B$

Lemma

特定
存在某個 $P_j \mid B = P_1 \times P_2 \cdots \times P_k + 1 \times 1$

$P_j \mid P_1 \times P_2 \times P_3$

$P_j \mid P_1 \times P_2 \times P_3 \times \cdots \times P_k \quad \times(-1)$

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$P_j \mid 1 \qquad P_j = \pm 1 \qquad$ (原本質數≥2)

$\times$

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$\boxed{\text{GCD 最大公因數}}$ $\qquad GCD(12, 18) = 6$

LCM 最小公倍數

<Def> $a \cdot b \in Z$

a positive integer $C$

We claim $c$ is GCD $\qquad$ a common divisor

if $c \mid a$ and $c \mid b$

$C_1 \cdot C_2, C_3 \cdots C_k$ ← 最大

greatest common divisor

$a \cdot b \in Z \qquad a \neq 0, \ b \neq 0$

$\exists c \in Z^+$ is GCD $\qquad 12, 18$

$2, 3, 6$

if (a) $c \mid a$, $c \mid b$ (最大)

(b) for any common divisor $d$ of $a$ and $b$

$2 \mid 6$ ✗

$3 \mid 6$ ✗

we have $d \mid c \qquad 6 \mid 6$ ✓

$2, 3, 6 \quad 6$

greatest maximum

least small minimum

<Thm> $a, b \in \mathbb{Z}^+$

there exist an unique $c \in \mathbb{Z}^+$ the gcd

<pf> Given $\overset{12}{a}, \overset{18}{b} \in \mathbb{Z}^+$ gcd 存在且唯一

$$S = \left\{ as+bt \;\middle|\; \begin{array}{l} s, t \in \mathbb{Z} \\ as+bt > 0 \end{array} \right\}$$

$S \neq \phi$

$S \subseteq \mathbb{Z}^+$

By the well-ordering property

$S$ has the least element $c$

We claim $c$ is the gcd 最大公因數是 a . b 兩數最小組合

$c \in S$ $c = ax+by$

$d \mid a$ $d \mid b$ 最大

$d \mid ax+by = c$

$d \mid c$

II If $c \nmid a$ $a = q_1 c + r$

$r = a - q_1 c$ 公因數

$0 < r < c$

$r = a - q_1 c = a - q_1(ax+by)$

$\quad = (1-q_1 x)a + (-q_1 y)b$

$r \in S$ ※ ($\because r < c$, $c$原本應為最小)

$c \nmid b$ $c \mid b$ ※

III. the

$c_1$ $c_2$ $c_1 \mid c_2$ $c_1 = c_2$ 唯一

$c_2 \mid c_1$